Engineering Mathematics (AHT-003) - Unit 2

Unit 2 - Calculus II - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers (syllabus): definite integrals and their properties, curve tracing, double and triple integrals, change of order and change of variables, Beta and Gamma functions.

Key ideas: Definite-integral properties save work: $\int_a^a f = \int_a^a f(a+b-x)$, and $\int_{-e}^{e} f = 2\int_0^e f$ for even f (0 for odd f). **Curve tracing**: check symmetry, origin, intercepts, asymptotes, tangents at origin before sketching - list the steps in the answer. **Double integrals** compute area/volume: integrate inner first, keeping the outer variable fixed. **Change of order**: redraw the region from the limits and rewrite with axes swapped - the sketch is everything. **Change of variables:** multiply by the Jacobian $|J|$ (polar: $dx\,dy = r\,dr\,d\theta$). **Beta/Gamma:** $\Gamma(n) = (n-1)!$, $\Gamma(1/2) = \sqrt{\pi}$, and $B(m,n) = \Gamma(m)\Gamma(n)/\Gamma(m+n)$ - they convert hard integrals into one-line answers.

Exam pointers: one "evaluate the double integral" (often needing order change - draw the region!), one Beta/Gamma evaluation using $\Gamma(1/2) = \sqrt{\pi}$, and a curve-tracing question are the regulars.